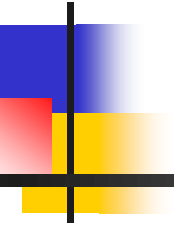
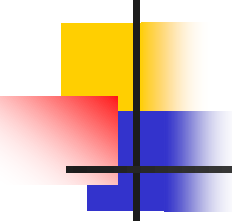


Satellite Communications



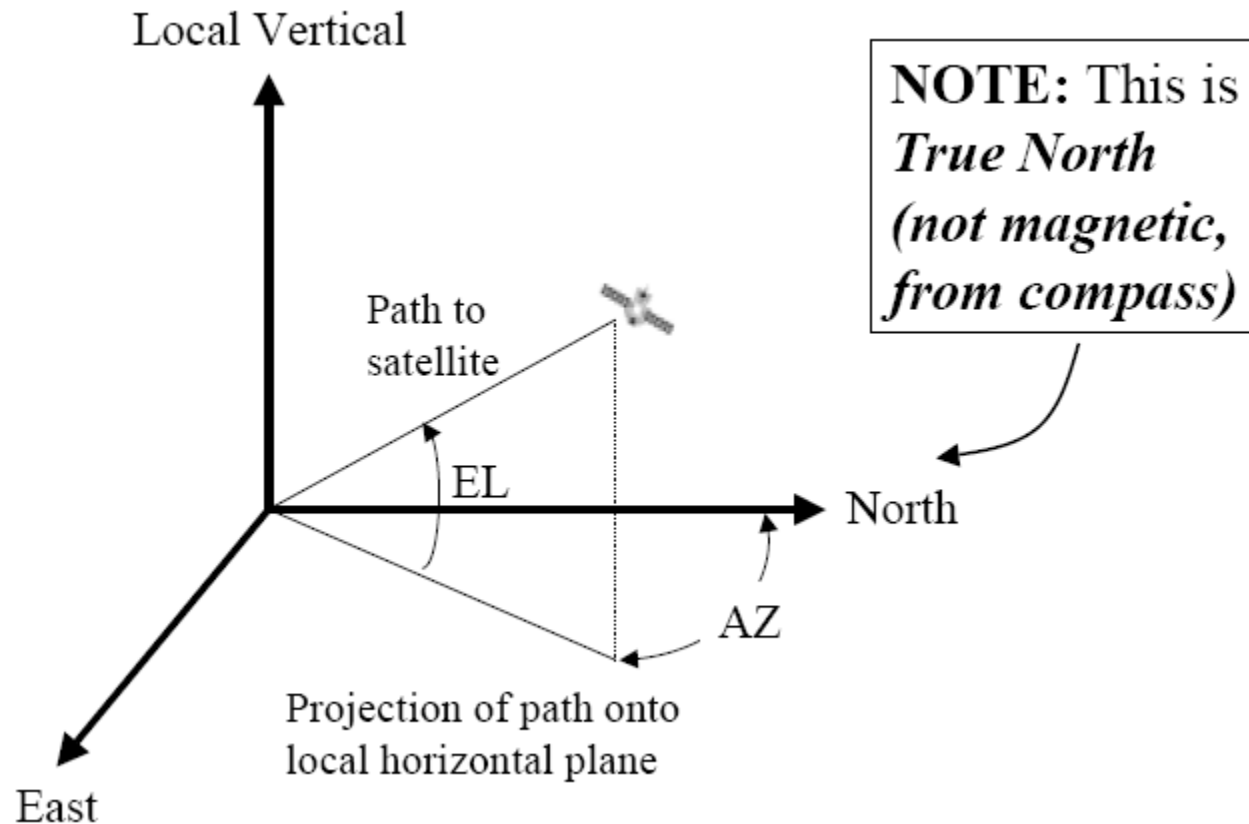
Look Angle Determination



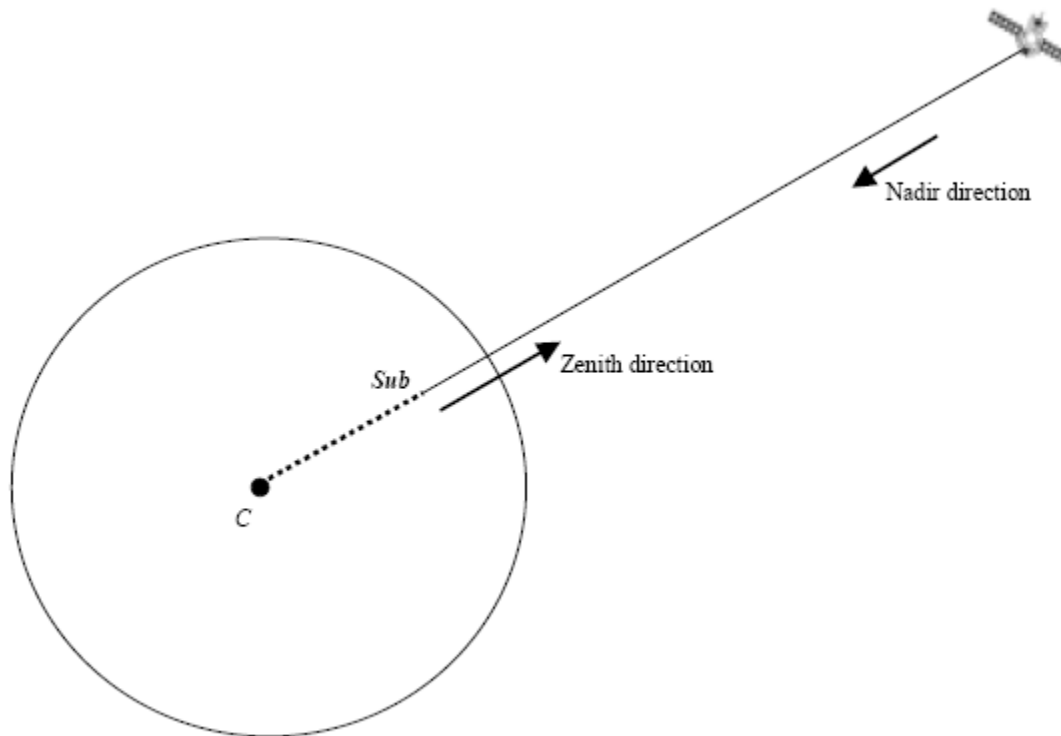
Look Angle Determination

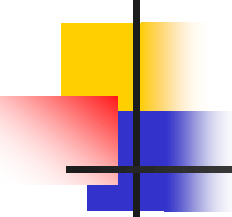
- Look angles: The coordinates to which an ES must point to communicate with a satellite. These are *azimuth* (AZ) and *elevation angle* (EL)
 - AZ: The angle measured from N to E to projection of satellite path onto horizontal plane
 - EL: The angle measured from the horizontal plane to the orbit plane
- The *subsattellite point*: The point, on the earth's surface, of intersection between a line from the earth's center to the satellite

Look Angle Definition



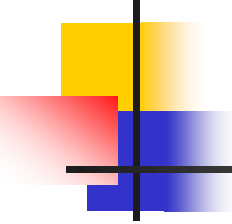
Definitions (Contd.)





Calculating Look Angle

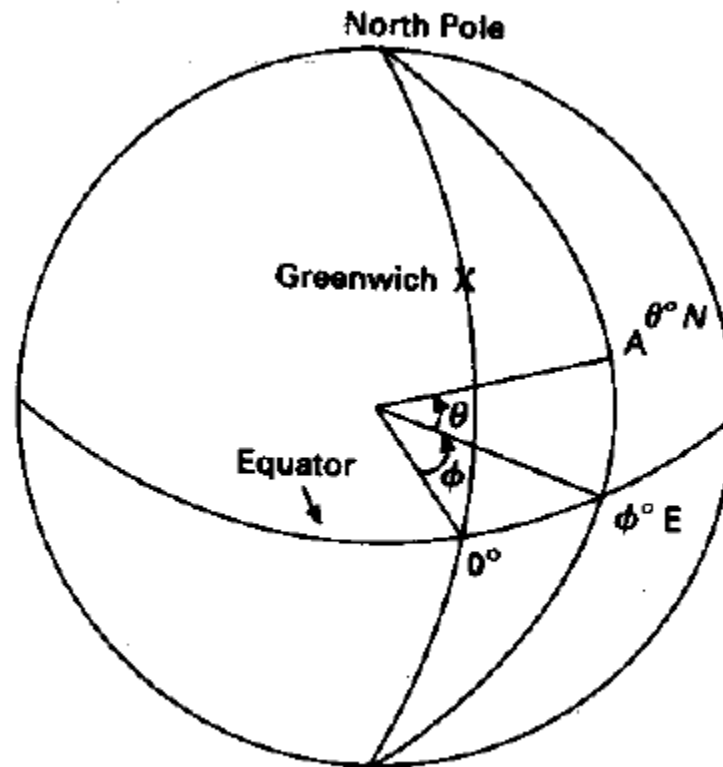
- *Need* six Orbital Elements
- *Calculate* the orbit from these Orbital Elements
- *Define* the orbital plane
- *Locate* satellite at time t with respect to the *First Point of Aries*
- *Find location* of the Greenwich Meridian relative to the first point of Aries
- *Use Spherical Trigonometry* to find the position of the satellite relative to a point on the earth's surface



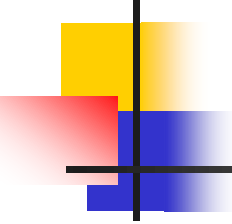
Coordinate System

- **Latitude:** Angular distance, measured in degrees, north or south of the equator.
 L from -90 to +90 (or from 90S to 90N)
- **Longitude:** Angular distance, measured in degrees, from a given reference longitudinal line (Greenwich, London).
 l from 0 to 360E (or 180W to 180E)

Coordinate System



Latitude ($\theta^{\circ}N$) and longitude ($\phi^{\circ}E$) of a point A.



Satellite Coordinates

- SUB-SATELLITE POINT
Latitude L_s
Longitude l_s
- EARTH STATION LOCATION
Latitude L_e
Longitude l_e
- Calculate γ , ANGLE AT EARTH CENTER
Between the line that connects the earth-center to the satellite
and the line from the earth-center to the earth station.

Geometry of Elevation Angle

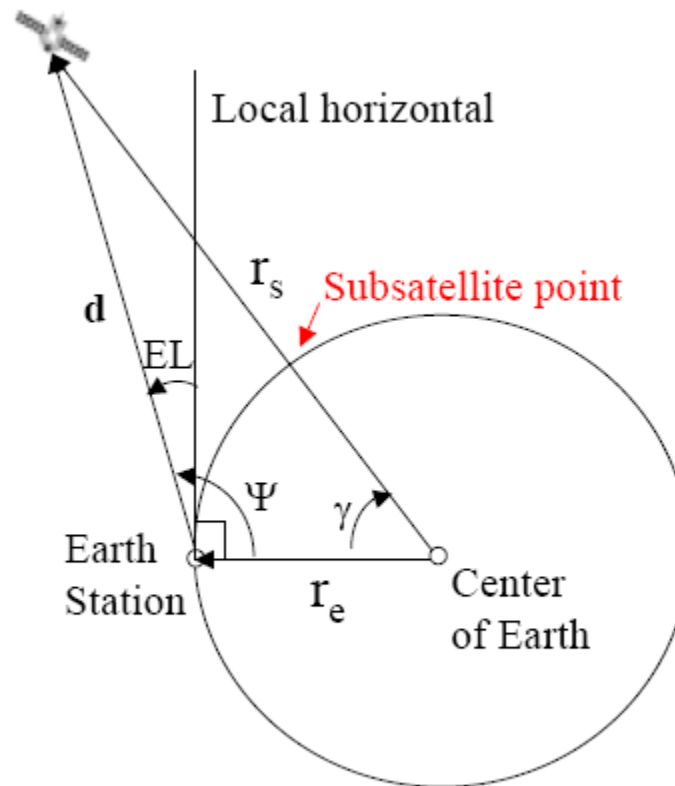
- Plane in picture is the one that includes center of the earth, Earth Station and Satellite.
- Subsatellite point will also be on the same plane.

$$El = \psi - 90^\circ$$

γ = central angle

r_s = radius to the satellite

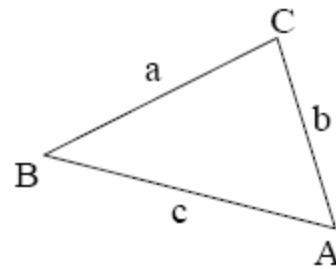
r_e = radius of the earth



Review of Geometry

- Review of plane trigonometry

- Law of Sines
- Law of Cosines
- Law of Tangents



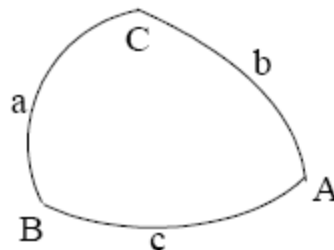
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\tan \frac{C}{2} = \sqrt{\frac{(d-a)(d-b)}{d(d-c)}}, d = \frac{a+b+c}{2}$$

- Review of spherical trigonometry

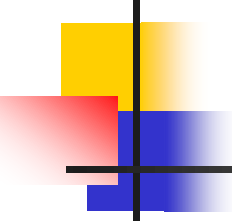
- Law of Sines
- Law of Cosines for angles
- Law of Cosines for sides



$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$



Central Angle

γ is defined so that it is non-negative and

$$\cos(\gamma) = \cos(L_e) \cos(L_s) \cos(l_s - l_e) + \sin(L_e) \sin(L_s)$$

The magnitude of the vectors joining the center of the earth, the satellite and the earth station are related by the law of cosine:

$$d = r_s \left[1 + \left(\frac{r_e}{r_s} \right)^2 - 2 \left(\frac{r_e}{r_s} \right) \cos(\gamma) \right]^{1/2}$$



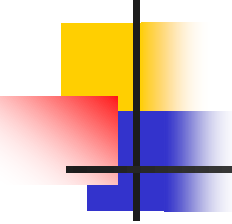
Elevation Angle Calculation

By the sine law we have

$$\frac{r_s}{\sin(\psi)} = \frac{d}{\sin(\gamma)}$$

Which yields

$$\cos(El) = \frac{\sin(\gamma)}{\left[1 + \left(\frac{r_e}{r_s}\right)^2 - 2\left(\frac{r_e}{r_s}\right)\cos(\gamma)\right]^{1/2}}$$



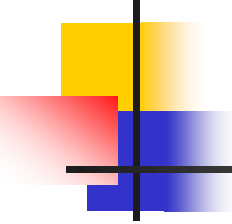
Example: Elevation Angle for GEO Satellite

Using $r_s = 42,164$ km and $r_e = 6,378.14$ km gives

$$d = 42,164 [1.0228826 - 0.3025396 \cos(\gamma)]^{1/2} \text{ km}$$

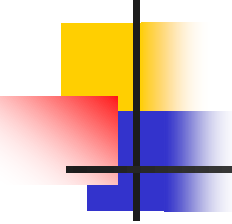
Which finally gives the elevation angle

$$\cos(El) = \frac{\sin(\gamma)}{[1.0228826 - 0.3025396 \cos(\gamma)]^{1/2}}$$



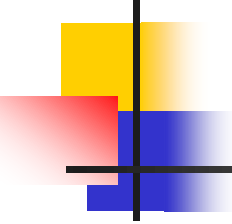
Azimuth Angle Calculation

- More complex approach for non-geo satellites.
- Different formulas and corrections apply depending on the combination of positions of the earth station and subsatellite point with relation to each of the four quadrants (NW, NE, SW, SE).
- Its calculation is simple for GEO satellites



Azimuth Angle Calculation for GEO Satellites

- SUB-SATELLITE POINT
 - Equatorial plane, Latitude **$L_s = 0^\circ$**
 - Longitude **l_s**
- EARTH STATION LOCATION
 - Latitude **L_e**
 - Longitude **l_e**



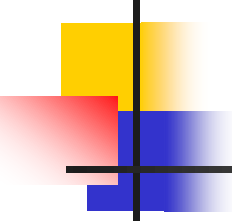
Azimuth Angle for GEO sat.

The original calculation previously shown:

$$\cos(\gamma) = \cos(L_e) \cos(L_s) \cos(l_s - l_e) + \sin(L_e) \sin(L_s)$$

Simplifies using $L_s = 0^\circ$ since the satellite is over the equator:

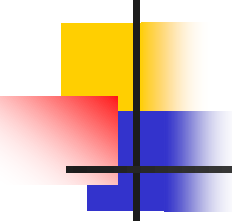
$$\cos(\gamma) = \cos(L_e) \cos(l_s - l_e)$$



Azimuth Angle for GEO sat.

To find the azimuth angle, an *intermediate angle*, α , must first be found. The intermediate angle allows the correct quadrant (see Figs. 2.10 & 2.13) to be found since the azimuthal direction can lie anywhere between 0° (true North) and clockwise through 360° (back to true North again). The intermediate angle is found from

$$\alpha = \tan^{-1} \left[\frac{\tan |l_s - l_e|}{\sin(L_e)} \right]$$



Azimuth Angle for GEO sat.

Case 1: Earth station in the Northern Hemisphere with

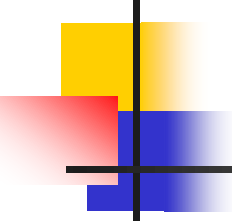
(a) Satellite to the SE of the earth station: $Az = 180^\circ - \alpha$

(b) Satellite to the SW of the earth station: $Az = 180^\circ + \alpha$

Case 2: Earth station in the Southern Hemisphere with

(c) Satellite to the NE of the earth station: $Az = \alpha$

(d) Satellite to the NW of the earth station: $Az = 360^\circ - \alpha$



Example for Look Angle Calculation of a GEO satellite

***FIND* the Elevation and Azimuth**

Look Angles for the following case:

Earth Station Latitude	52° N	}	London, England Dockland region
Earth Station Longitude	0°		
Satellite Latitude	0°	}	Geostationary INTELSAT IOR Primary
Satellite Longitude	66° E		



Example (Contd.)

Step 1. Find the central angle γ

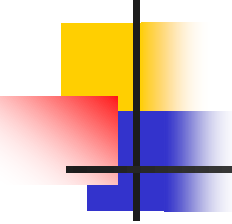
$$\begin{aligned}\cos(\gamma) &= \cos(L_e) \cos(l_s - l_e) \\ &= \cos(52) \cos(66) \\ &= 0.2504\end{aligned}$$

yielding $\gamma = 75.4981^\circ$

Step 2. Find the elevation angle El

$$\cos(El) = \frac{\sin(\gamma)}{[1.0228826 - 0.3025396 \cos(\gamma)]^{1/2}}$$

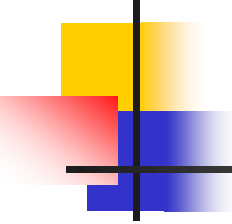
$$El = 5.85^\circ$$



Example (Contd.)

Step 3. Find the intermediate angle, α

$$\begin{aligned}\alpha &= \tan^{-1} \left[\frac{\tan |(l_s - l_e)|}{\sin (L_e)} \right] \\ &= \tan^{-1} [(\tan (66 - 0)) / \sin (52)] \\ &= 70.6668\end{aligned}$$



Example (Contd.)

The earth station is in the Northern hemisphere and the satellite is to the South East of the earth station. This gives

$$\begin{aligned} Az &= 180^\circ - \alpha \\ &= 180 - 70.6668 = 109.333^\circ \text{ (clockwise from true North)} \end{aligned}$$

ANSWER: The look-angles to the satellite are

$$\text{Elevation Angle} = 5.85^\circ$$

$$\text{Azimuth Angle} = 109.33^\circ$$